

VALUE INVESTING ENGINE

A Data-Driven Company Evaluation

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Data Analytics Bootcamp
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Why this project?

Investment decisions are often driven by emotions and market noise.

I wanted to build a Value-Investing tool that:

- Evaluates companies objectively by financial fundamentals
- Ranks and compares investment opportunities
- Automates financial analysis



Project Objectives

Main Objective

- Develop a simplified value investing framework capable of ranking publicly traded companies based on their financial quality.

Specific Objectives

- Extract financial data
- Calculate KPIs
- Create an investment scoring model
- Rank companies
- Analyse fundamentals as well as market behavior (“Mr. Market”)
- Build datasets & interactive dashboards

Data

Data Source from **Yahoo Finance**

- Financial statements (Income Statement, Cash Flow, Balance Sheet)
- Historical stock prices
- Benchmark data (S&P 500), Universe of ~500 american companies

Data Processing

- Python
- HTML Scraping with request
- Pandas
- NumPy

Dataset Characteristics

Companies analyzed: 27

Years analyzed: 2022,
2023, 2024, 2025

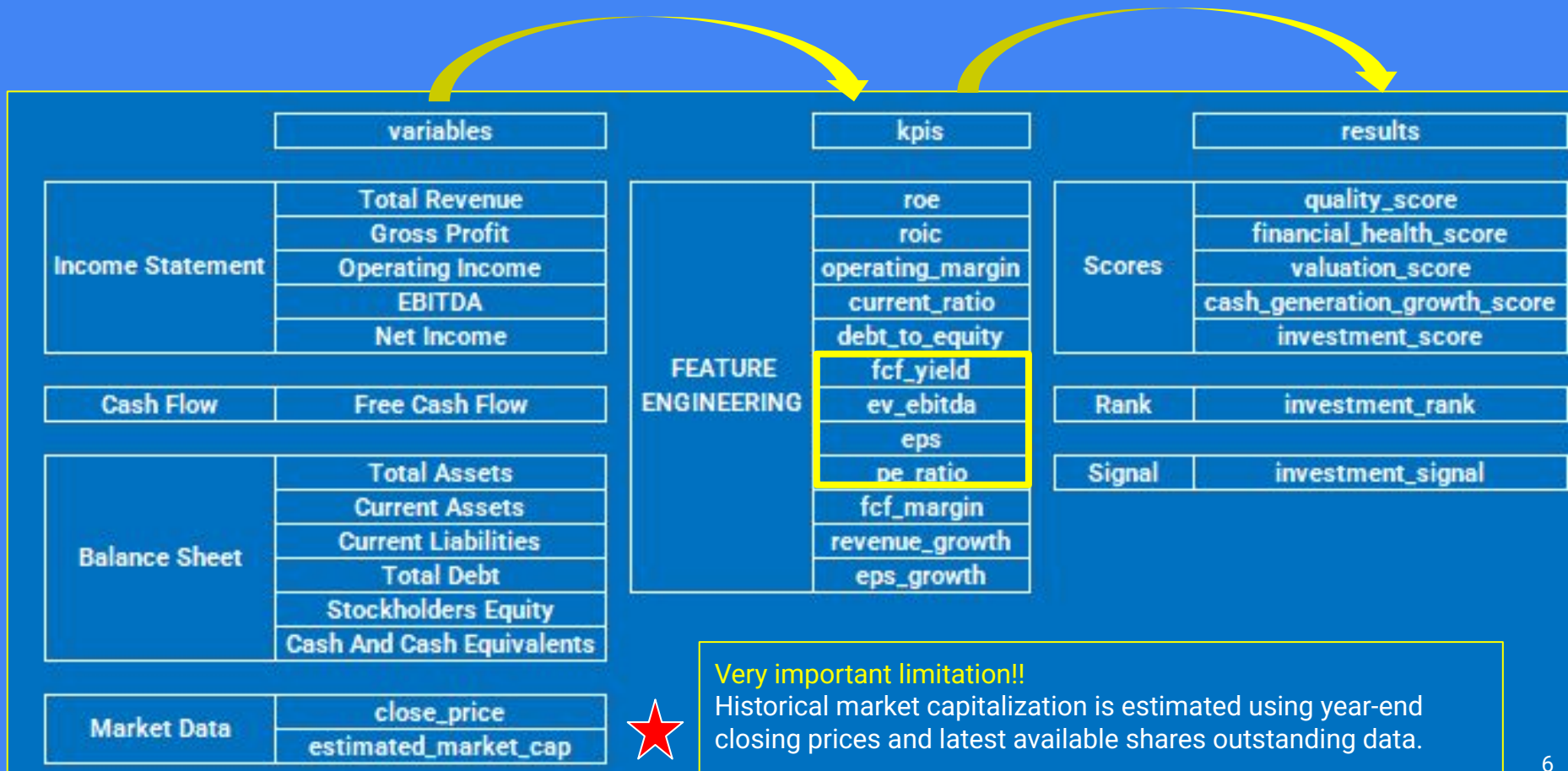
Final Records: 104

Feature Engineering & KPIs

Weighting	Dimension	KPIs
0.30	Profitability / Quality	$0.30 * ROE + 0.40 * ROIC + 0.30 * \text{Operating Margin}$
0.20	Financial Health	$0.40 * \text{Current Ratio} + 0.60 * \text{Debt-to-Equity}$
0.25	Valuation	$0.50 * \text{FCF Yield} + 0.30 * \text{EV/EBITDA} + 0.20 * \text{P/E}$
0.25	Cash Generation + Growth	$0.60 * \text{FCF Margin} + 0.20 * \text{Revenue Growth} + 0.20 * \text{EPS Growth}$

Each KPI was standardized into a score ranging from 0 to 1.

Process Flow Diagram



How does the Investment Score work?

Quality | Financial Health | Valuation | Cash Generation & Growth

↓
KPI Scores

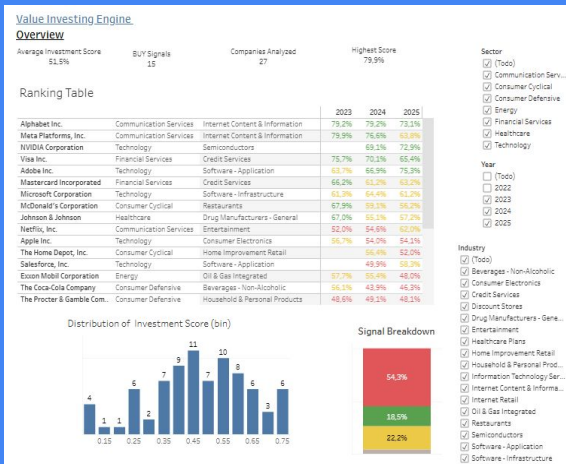
↓
Investment Score = Weighted Average(KPI Scores)

↓
INVESTMENT SIGNALS
BUY / HOLD / SELL

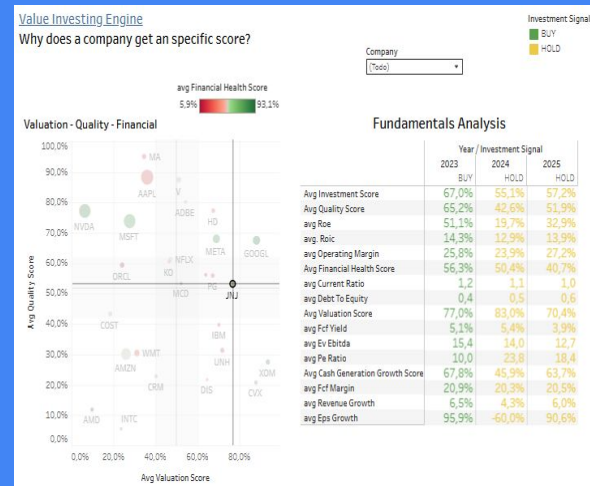
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INVESTMENT Ranking per year
from 1 to 27

Dashboard Structure

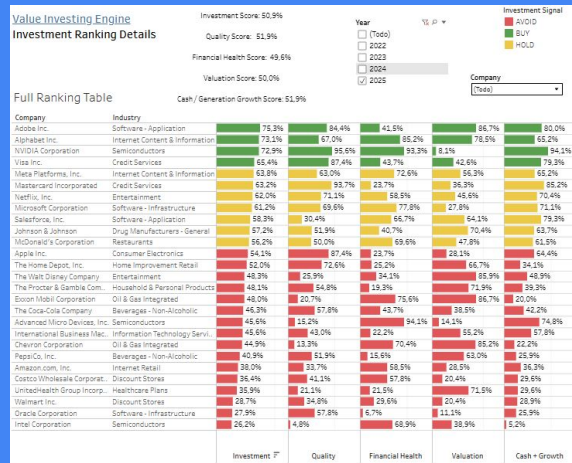
Dashboard 1 Overview



Dashboard 3 Fundamentals Analysis



Dashboard 2 Investment Ranking Details



Dashboard 4 “Mr. Market”

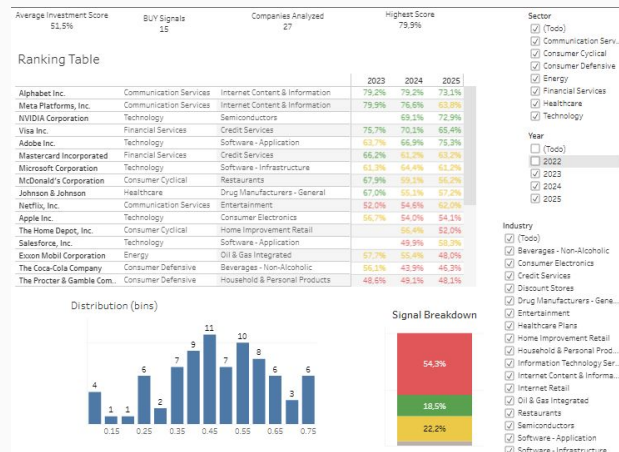


Live Dashboard Demonstration

From raw financial data to investment insights

Click Link

https://public.tableau.com/views/ValueInvestingEngine/Cover?:language=es-ES&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link



with interactive filters

In 2025, 4 companies received a **BUY signal**:

- ADBE | Adobe Inc. Technology Software - Application
- GOOGL | Alphabet Inc. Communication Services Internet Content & Information
- NVDA | NVIDIA Corporation Technology Semiconductors
- V | Visa Inc. Financial Services Credit Services

Representing almost

15%

of the analyzed universe.



Backtesting Performance of Investment Score Model (*)

(*) curated universe of large surviving companies; the backtest should be considered illustrative rather than a fully out-of-sample validation.

*Evaluation of BUY signals
using forward 1-year
returns*

Average BUY Return	BUY Success Rate
22.9%	72.7% (8 / 11)

*Directionally encouraging
on a small sample, not a
statistical proof.*

	ticker	year	investment_score	investment_signal	close_price	company	sector	industry	next_close	future_return
63	META	2023	0.799034	BUY	351.203949	Meta Platforms, Inc.	Communication Services	Internet Content & Information	583.170227	0.6605
33	GOOGL	2024	0.791852	BUY	188.428864	Alphabet Inc.	Communication Services	Internet Content & Information	312.779816	0.6599
75	NVDA	2024	0.690926	BUY	134.089722	NVIDIA Corporation	Technology	Semiconductors	186.272797	0.3892
32	GOOGL	2023	0.792110	BUY	138.544922	Alphabet Inc.	Communication Services	Internet Content & Information	188.428864	0.3601
55	MA	2023	0.661506	BUY	420.279877	Mastercard Incorporated	Financial Services	Credit Services	521.879700	0.2417
94	V	2023	0.756691	BUY	255.495255	Visa Inc.	Financial Services	Credit Services	312.517487	0.2232
64	META	2024	0.763889	BUY	583.170227	Meta Platforms, Inc.	Communication Services	Internet Content & Information	659.525330	0.1309
95	V	2024	0.701111	BUY	312.517487	Visa Inc.	Financial Services	Credit Services	349.264221	0.1176
59	MCD	2023	0.676779	BUY	279.429382	McDonald's Corporation	Consumer Cyclical	Restaurants	279.811157	0.0014
47	JNJ	2023	0.670306	BUY	145.799454	Johnson & Johnson	Healthcare	Drug Manufacturers - General	138.784348	-0.0481
6	ADBE	2024	0.669444	BUY	444.679993	Adobe Inc.	Technology	Software - Application	349.989990	-0.2129

Challenges & Lessons Learned

Main Challenges

- Scaling an ambitious idea to a realistic scope
- Working with limited historical data and some missing values.
- Learning Tableau from scratch
- Balancing analytical depth with with simple visuals

Key Lesson

I developed technical (Python, Tableau, GitHub) and business skills (Financial Analysis, KPI Design, Data Storytelling). More importantly, I learned that building dashboards is about turning complex data into clear, **actionable insights**.

A successful data project is not about building everything you imagine, but about delivering meaningful insights with the data, tools, and constraints available.

Future Improvements

- Use professional data APIs (e.g., Alpha Vantage) for more reliable data.
- Expand the universe of companies (multiple sectors and industries) for robust comparisons. With $n = 27$, "normalization from 0 to 1" depends a lot on which one is in the sample.
- Add a Discounted Cash Flow (DCF) module to estimate intrinsic value and margin of safety (*).

(*) Concepts derived from prominent investors such as Graham, Buffett, and Damodaran

"Price is what you pay. Value is what you get"



Warren Buffett

Questions?

Thank you
for your
attention!

Julieta Mirensky

